

Olympiad Test Paper — Science (Class 7)

Chapter 10: Respiration in Organisms

Subject	Science (NCERT Class 7)
Chapter	10 — Respiration in Organisms
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: +4 for each correct answer · -1 for each wrong answer · 0 if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. Remember the word equation for cellular respiration: glucose + oxygen → carbon dioxide + water + energy. Keep "breathing" (the exchange of air) separate from "respiration" (the release of energy inside cells). No calculator is needed. Most questions need two to four reasoning steps; watch for traps. Do not write on this paper — mark answers on your answer sheet.

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- During respiration, the food broken down to release energy in our cells is mainly:
(A) protein (B) fat
(C) glucose (D) vitamins
 - Which statement correctly separates breathing from respiration?
(A) breathing releases energy; respiration exchanges gases
(B) breathing is the exchange of gases; respiration releases energy inside cells
(C) breathing and respiration mean exactly the same thing
(D) respiration happens only in the lungs, breathing in the cells
 - A jar of exhaled air is bubbled through clear limewater. What is observed, and why?
(A) it stays clear, because exhaled air has no carbon dioxide
(B) it turns blue, because of extra oxygen (C) it turns red, because of water vapour
(D) it turns milky, because exhaled air is rich in carbon dioxide
 - The correct word equation for aerobic respiration is:
(A) glucose + oxygen → carbon dioxide + water + energy
(B) glucose → alcohol + carbon dioxide + energy
(C) carbon dioxide + water → glucose + oxygen
(D) glucose + carbon dioxide → oxygen + water + energy
 - A yeast cell respiring in dough without oxygen produces:

(A) lactic acid and energy

(B) carbon dioxide and water only

(C) oxygen and glucose

(D) alcohol, carbon dioxide and energy

6. After sprinting hard, an athlete's leg muscles ache and cramp. The substance responsible is:

(A) lactic acid

(B) alcohol

(C) carbon dioxide

(D) glucose

7. Anaerobic respiration is best described as the breakdown of glucose:

(A) only inside the lungs

(B) using large amounts of oxygen

(C) in the absence of oxygen

(D) only in green plants

8. During inhalation, the diaphragm:

(A) flattens and moves down, and the chest cavity gets larger

(B) domes up, and the chest cavity gets smaller

(C) stays still while only the ribs move

(D) flattens and moves down, and the chest cavity gets smaller

9. The correct path of air entering the human body is:

(A) nostrils → lungs → trachea → nasal cavity

(B) trachea → nostrils → nasal cavity → lungs

(C) nostrils → nasal cavity → trachea → lungs

(D) nasal cavity → nostrils → lungs → trachea

10. Insects such as cockroaches take in air through small openings called:

(A) spiracles

(B) gills

(C) stomata

(D) lenticels

11. A fish kept in a sealed bag of boiled (air-free) water soon dies. The best reason is that:

(A) gills extract dissolved oxygen from water, and boiled water has lost it

(B) fish breathe by swallowing water into their lungs

(C) fish need carbon dioxide dissolved in the water

(D) boiled water is too heavy for the gills to pump

12. A person breathes 18 times per minute at rest. About how many breaths is that in 5 minutes?

(A) 36

(B) 23

(C) 360

(D) 90

13. Which organ takes in oxygen in an earthworm?

(A) gills

(B) its moist skin

(C) lungs

(D) spiracles

14. The part of the respiratory system commonly called the windpipe is the:

(A) trachea

(B) diaphragm

(C) larynx

(D) bronchus

- 15.** During heavy exercise our breathing rate rises mainly because:
- (A) the lungs shrink and need refilling
 - (B) muscles need more oxygen and more carbon dioxide must be removed
 - (C) the diaphragm has stopped working
 - (D) the body is trying to cool the blood
- 16.** A frog can take in oxygen through:
- (A) gills only
 - (B) spiracles only
 - (C) stomata in its skin
 - (D) both its moist skin and its lungs
- 17.** In plants, the tiny pores on the surface of leaves used for gas exchange are the:
- (A) lenticels
 - (B) spiracles
 - (C) alveoli
 - (D) stomata
- 18.** The breakdown of glucose with the release of energy takes place:
- (A) only in the lungs
 - (B) in the cells of the body
 - (C) only in the blood
 - (D) only in the nostrils
- 19.** Which of these is the correct comparison of the energy released per glucose molecule?
- (A) anaerobic respiration releases more energy than aerobic respiration
 - (B) aerobic respiration releases more energy than anaerobic respiration
 - (C) both release exactly the same energy
 - (D) anaerobic respiration releases no energy at all
- 20.** The muscle that lies below the lungs and helps in breathing is the:
- (A) diaphragm
 - (B) trachea
 - (C) bronchus
 - (D) heart
- 21.** A muscle cramp after running is usually relieved by:
- (A) running even faster
 - (B) holding your breath
 - (C) rest, massage or a hot bath
 - (D) drinking less water
- 22.** Compared with breathing in air, a fish must pump water over its gills almost continuously because:
- (A) water has no oxygen at all
 - (B) gills cannot move on their own
 - (C) fish do not respire
 - (D) water contains far less dissolved oxygen than air does
- 23.** During exhalation, the ribs and the chest cavity respectively:
- (A) move up and out; cavity gets larger
 - (B) move down and in; cavity gets smaller
 - (C) stay still; cavity gets larger
 - (D) move down and in; cavity gets larger
- 24.** Which set lists three organs/structures used for taking in oxygen by three DIFFERENT organisms?
- (A) gills (human), lungs (fish), spiracles (earthworm)
 - (B) stomata (fish), lungs (earthworm), gills (human)

- (C) spiracles (fish), gills (cockroach), lungs (frog)
- (D) gills (fish), lungs (human), moist skin (earthworm)

25. When you exhale onto a cold mirror, it fogs up. This shows exhaled air contains more:

- (A) oxygen
- (B) water vapour
- (C) nitrogen
- (D) glucose

26. A swimmer's breathing rate is measured as 15 per minute before a race and 35 per minute just after. The rise happens because:

- (A) the lungs have become smaller
- (B) the heart has stopped beating
- (C) the body needs to supply more oxygen and remove more carbon dioxide
- (D) less energy is needed during the race

27. Yeast is used to make bread rise. The gas that forms the bubbles in the dough is:

- (A) oxygen
- (B) water vapour
- (C) carbon dioxide
- (D) nitrogen

28. Roots of a plant obtain oxygen for respiration:

- (A) from sunlight falling on the leaves
- (B) from air present in the spaces between soil particles
- (C) from carbon dioxide in the air
- (D) only from water in the soil

29. Which statement about breathing rate is correct?

- (A) it is lower at rest and rises during exercise
- (B) it is highest while sleeping and lowest while running
- (C) it stays exactly the same whatever you do
- (D) it only changes if you hold your breath

30. The thin-walled, balloon-like sacs in the lungs where oxygen passes into the blood are the:

- (A) bronchi
- (B) alveoli
- (C) spiracles
- (D) bronchioles

31. A plant carries out respiration:

- (A) both day and night
- (B) only during the day
- (C) only at night
- (D) never, because it makes its own food

32. Which is the most complete statement about gas exchange in respiration?

- (A) we take in carbon dioxide and give out oxygen
- (B) we take in and give out only oxygen
- (C) we take in oxygen and give out carbon dioxide
- (D) we take in nitrogen and give out carbon dioxide

33. In the stem of a woody plant, gas exchange takes place through openings called:

- (A) stomata
- (B) spiracles
- (C) alveoli
- (D) lenticels

- 34.** The air we breathe out, compared with the air we breathe in, has:
 (A) more oxygen and less carbon dioxide (B) more oxygen and more carbon dioxide
 (C) the same amounts of both gases (D) less oxygen and more carbon dioxide
- 35.** During the rising of bread dough, yeast is carrying out:
 (A) anaerobic respiration, producing alcohol and carbon dioxide
 (B) aerobic respiration, producing only water (C) photosynthesis, producing glucose
 (D) breathing through tiny lungs
- 36.** Cockroaches and other insects carry air to their body tissues through a network of tubes called:
 (A) bronchi (B) gills
 (C) alveoli (D) tracheae
- 37.** Which sequence shows the breathing movements in the correct order during one inhalation?
 (A) diaphragm contracts → ribs move up and out → chest expands → air rushes in
 (B) diaphragm relaxes → ribs move down → chest shrinks → air goes out
 (C) ribs move down → diaphragm domes up → chest expands → air rushes in
 (D) air rushes in → diaphragm relaxes → ribs move down → chest shrinks
- 38.** A burning candle and a respiring animal both:
 (A) release oxygen and use carbon dioxide (B) need no oxygen at all
 (C) use oxygen and release energy and carbon dioxide
 (D) produce only water and no gases
- 39.** Which organism does NOT use lungs to take in oxygen?
 (A) human (B) frog
 (C) fish (D) dog
- 40.** After a fizzy fermentation experiment, a candle lowered into the gas above the liquid goes out. The gas is most likely:
 (A) oxygen (B) hydrogen
 (C) nitrogen (D) carbon dioxide
- 41.** The number of breaths a person takes in one minute is called the:
 (A) heart rate (B) pulse
 (C) breathing rate (D) respiration product
- 42.** Mountaineers at very high altitude often feel breathless. The best Class-7 reason is that:
 (A) the air there has more carbon dioxide (B) their lungs stop working in the cold
 (C) the air there is thinner, so each breath supplies less oxygen
 (D) there is no air at all on mountains

43. Which pairing of organism and breathing organ is INCORRECT?

- (A) cockroach — gills
- (B) fish — gills
- (C) earthworm — moist skin
- (D) human — lungs

44. During vigorous exercise, when oxygen supply to muscles cannot keep up, the muscles partly switch to:

- (A) photosynthesis
- (B) breathing through the skin
- (C) aerobic respiration only
- (D) anaerobic respiration, forming lactic acid

45. A balloon attached to the mouth of a flask of sugar solution with yeast slowly inflates. The gas inflating it is:

- (A) oxygen made by the yeast
- (B) water vapour
- (C) nitrogen from the air
- (D) carbon dioxide from anaerobic respiration

46. Which statement is TRUE?

- (A) every living cell, plant or animal, respire
- (B) only animals respire; plants only photosynthesise
- (C) respiration stops in animals while they sleep
- (D) only the lungs respire in a human

47. Two test tubes of limewater are set up: tube X receives air breathed OUT, tube Y receives ordinary room air pushed in by a pump for the same time. Which is correct?

- (A) Y turns milky much faster than X
- (B) X turns milky much faster than Y
- (C) neither changes at all
- (D) both turn milky equally fast

48. The voice box, located in the throat above the windpipe, is the:

- (A) bronchus
- (B) diaphragm
- (C) alveolus
- (D) larynx

49. A student says, "Fish breathe water." The most accurate correction is:

- (A) fish do breathe water directly into their lungs
- (B) fish make their own oxygen from water
- (C) fish take in oxygen dissolved in water using their gills
- (D) fish do not need oxygen at all

50. At rest a person breathes about 15–18 times per minute. Which value would you most expect just after climbing several flights of stairs?

- (A) 5 per minute
- (B) 16 per minute
- (C) 30 per minute
- (D) 0 per minute

51. Which best explains why air rushes INTO the lungs during inhalation?

- (A) the lungs pull the air in like a magnet
- (B) the chest cavity expands, lowering the pressure inside the lungs

- (C) the diaphragm pushes the air in from below
- (D) the air is heavier than the lungs

52. Earthworms come to the surface of waterlogged soil after heavy rain mainly because:

- (A) they want to drink the rainwater
- (B) the water fills the soil spaces, leaving too little air for skin breathing
- (C) the soil becomes too cold for them
- (D) they breathe better when fully underwater

53. A green plant in a sealed dark box for a night will:

- (A) release oxygen and use carbon dioxide
- (B) neither take in nor give out any gas
- (C) use oxygen and release carbon dioxide through respiration
- (D) carry out photosynthesis as usual

54. Which of these is a product common to BOTH aerobic respiration and the fermentation of sugar by yeast?

- (A) carbon dioxide
- (B) lactic acid
- (C) oxygen
- (D) alcohol

55. The main job of the hair and mucus inside the nasal cavity is to:

- (A) add oxygen to the air
- (B) trap dust and germs and clean the incoming air
- (C) cool the lungs down
- (D) produce energy

56. During exhalation, air leaves the lungs because:

- (A) the chest cavity expands and pressure drops
- (B) the ribs move further up and out
- (C) the diaphragm relaxes and the chest cavity shrinks, raising the pressure inside
- (D) the lungs make extra oxygen that must escape

57. A jar holds a small germinating-seed setup that uses up the oxygen; a candle lowered in goes out and limewater turns milky. This shows the seeds were:

- (A) photosynthesising, giving out oxygen
- (B) respiring, using oxygen and giving out carbon dioxide
- (C) not alive
- (D) giving out only water vapour

58. Which statement comparing breathing in different animals is correct?

- (A) a cockroach uses gills while a fish uses spiracles
- (B) a fish uses gills while a cockroach uses tracheae
- (C) an earthworm uses lungs while a human uses skin
- (D) a frog uses spiracles while an earthworm uses gills

59. A bicycle pump and the human breathing system are alike because both:

- (A) change the volume of a space to move air in and out
- (B) burn glucose to release energy
- (C) make oxygen from water

(D) turn limewater milky

60. Why must even a sleeping person keep breathing, although the muscles are barely moving?

(A) cells stop respiring during sleep, so air must be stored

(B) breathing only happens during exercise

(C) the lungs need air to make glucose at night

(D) every living cell keeps respiring and needs a constant oxygen supply

End of question paper. Maximum marks: 240. Marking: +4 correct, -1 wrong, 0 unattempted.

Solutions — Science (Class 7), Chapter 10: Respiration in Organisms

Marking: +4 correct, –1 wrong, 0 unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	C	11	A	21	C	31	A	41	C	51	B
2	B	12	D	22	D	32	C	42	C	52	B
3	D	13	B	23	B	33	D	43	A	53	C
4	A	14	A	24	D	34	D	44	D	54	A
5	D	15	B	25	B	35	A	45	D	55	B
6	A	16	D	26	C	36	D	46	A	56	C
7	C	17	D	27	C	37	A	47	B	57	B
8	A	18	B	28	B	38	C	48	D	58	B
9	C	19	B	29	A	39	C	49	C	59	A
10	A	20	A	30	B	40	D	50	C	60	D

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

- (C)** Respiration breaks down food to release energy, and the chief fuel is glucose (a simple sugar). Proteins and fats can be used in special situations, but the everyday fuel of the cell is glucose.
- (B)** Breathing is only the physical exchange of air (taking in oxygen-rich air, giving out carbon-dioxide-rich air). Respiration is the chemical process inside the cells that breaks down glucose to release energy. They are not the same — breathing supplies the oxygen that respiration uses.
- (D)** Carbon dioxide turns limewater milky. Exhaled air is rich in CO₂, so the limewater goes milky. ("Stays clear" wrongly assumes exhaled air has no CO₂.)
- (A)** Aerobic respiration: glucose + oxygen → carbon dioxide + water + energy. Option (C) is photosynthesis written backwards, and (B) is the anaerobic (yeast) equation.
- (D)** Without oxygen (anaerobic), yeast carries out fermentation: glucose → alcohol + carbon dioxide + energy. Lactic acid is what *our muscle* cells make, not yeast.

- 6. (A)** When muscles run short of oxygen during a sprint, they respire anaerobically and make lactic acid, which builds up and causes the ache and cramp.
- 7. (C)** "An-aerobic" literally means "without air (oxygen)". So anaerobic respiration is the breakdown of glucose in the absence of oxygen.
- 8. (A)** On inhalation the diaphragm contracts, flattens and moves down; together with the ribs swinging up and out, this enlarges the chest cavity so air flows in. (Option D pairs the right diaphragm move with the wrong volume change — a trap.)
- 9. (C)** Air travels: nostrils → nasal cavity → trachea (windpipe) → lungs. The other orders scramble this path.
- 10. (A)** Insects such as cockroaches take in air through tiny holes called spiracles. (Gills are for fish, stomata and lenticels are for plants.)
- 11. (A)** Fish do not have lungs; their gills extract oxygen *dissolved* in the water. Boiling drives the dissolved oxygen out, so the fish suffocates even though it is surrounded by water.
- 12. (D)** $18 \text{ breaths/minute} \times 5 \text{ minutes} = 90 \text{ breaths}$. (36 would be 18×2 ; 360 multiplies by 20 by mistake.)
- 13. (B)** An earthworm has no lungs or gills; it takes in oxygen through its moist skin, which is why it must stay damp.
- 14. (A)** The trachea is the windpipe — the tube carrying air from the throat toward the lungs.
- 15. (B)** Hard exercise makes muscles respire faster, so they need more oxygen and produce more carbon dioxide. Breathing speeds up to supply the extra O_2 and clear the extra CO_2 .
- 16. (D)** A frog is special: it can take in oxygen both through its moist skin and through its lungs.
- 17. (D)** Leaves exchange gases through tiny pores called stomata. (Lenticels are on stems, spiracles on insects, alveoli in human lungs.)
- 18. (B)** Respiration — the actual breakdown of glucose to release energy — happens inside the body's cells, not in the lungs or blood (those only transport the gases).
- 19. (B)** Aerobic respiration breaks glucose down completely, releasing much more energy per glucose molecule than anaerobic respiration, which only breaks it down partly (into alcohol or lactic acid).

- 20. (A)** The diaphragm is the dome-shaped sheet of muscle below the lungs that drives breathing movements.
- 21. (C)** Lactic-acid cramps ease once oxygen returns and the acid is cleared; rest, gentle massage or a hot bath all help, so the body can finish breaking down the lactic acid.
- 22. (D)** Water holds far less dissolved oxygen than the same volume of air, so a fish must keep passing a steady stream of water over its gills to gather enough oxygen.
- 23. (B)** On exhalation the diaphragm relaxes (domes up) and the ribs move down and in, so the chest cavity gets smaller and air is pushed out.
- 24. (D)** A correct set must match each organ to the right organism: gills (fish), lungs (human), moist skin (earthworm). The other sets mismatch organs and organisms.
- 25. (B)** The mirror fogs because exhaled air carries extra water vapour, which condenses on the cold glass.
- 26. (C)** Swimming hard makes the muscles respire faster, so the body must take in more oxygen and remove more carbon dioxide — hence the jump from 15 to 35 breaths per minute.
- 27. (C)** Yeast fermenting in dough gives off carbon dioxide; the trapped bubbles of CO₂ make the bread rise.
- 28. (B)** Soil has air in the spaces between its particles; roots take in oxygen from this trapped air for their own respiration.
- 29. (A)** Breathing rate is lowest at rest and rises during exercise (the opposite of option B, which is the trap).
- 30. (B)** The tiny balloon-like air sacs in the lungs are the alveoli; their thin walls let oxygen pass into the surrounding blood.
- 31. (A)** Plants respire both day and night. (They photosynthesise only in light, but respiration never stops — the trap is thinking plants only respire at night.)
- 32. (C)** In respiration we take in oxygen and give out carbon dioxide. (Taking in CO₂ and giving out O₂ is photosynthesis in plants — a common mix-up.)
- 33. (D)** Woody stems exchange gases through small openings called lenticels. (Stomata are on leaves.)
- 34. (D)** Compared with inhaled air, exhaled air has less oxygen (some was used up) and more carbon dioxide (a product of respiration).

- 35. (A)** In the dough, yeast lacks oxygen and respire anaerobically, producing alcohol and carbon dioxide — the CO₂ makes the dough rise.
- 36. (D)** From the spiracles, air travels through a branching network of fine air tubes called tracheae that carry it directly to the insect's tissues.
- 37. (A)** One inhalation, in order: diaphragm contracts → ribs move up and out → chest expands → air rushes in. The other orders describe exhalation or scramble the steps.
- 38. (C)** A burning candle and a respiring animal both take in oxygen and release energy plus carbon dioxide — respiration is like a slow, controlled "burning" of glucose.
- 39. (C)** A fish does not have lungs; it uses gills. Humans, frogs and dogs all use lungs (a frog also uses its skin), so the fish is the odd one out.
- 40. (D)** A flame needs oxygen; if the candle goes out, the gas does not support burning. Fermentation gives off carbon dioxide, which puts the flame out.
- 41. (C)** The number of breaths taken in one minute is the breathing rate. (Heart rate and pulse count heartbeats, not breaths.)
- 42. (C)** High up, the air is thinner (lower pressure), so each breath carries less oxygen; the body feels short of oxygen and breathing speeds up — that breathless feeling.
- 43. (A)** The incorrect pair is "cockroach — gills". A cockroach breathes through spiracles and tracheae, not gills. The other three pairings are all correct.
- 44. (D)** When oxygen can't reach the muscles fast enough during vigorous exercise, they switch partly to anaerobic respiration, forming lactic acid.
- 45. (D)** Yeast in sugar solution respire anaerobically, releasing carbon dioxide that collects in and inflates the balloon. (Yeast does not *make* oxygen.)
- 46. (A)** Every living cell — in plants as well as animals, awake or asleep — respire to release the energy it needs to stay alive.
- 47. (B)** Tube X receives *exhaled* air (CO₂-rich) and tube Y receives ordinary room air. The exhaled-air tube (X) turns milky far faster because exhaled air carries much more carbon dioxide. So "X turns milky much faster than Y" is the correct statement.
- 48. (D)** The voice box sitting at the top of the windpipe is the larynx.
- 49. (C)** The accurate correction to "fish breathe water" is that fish take in oxygen *dissolved* in water using their gills — they do not breathe water into lungs, nor make their own oxygen.

- 50. (C)** Resting rate is about 15–18 per minute; after climbing stairs it rises well above that, so ~30 per minute is the expected value. (16 is a resting value; 0 and 5 are impossible after exertion.)
- 51. (B)** When the chest cavity expands, the same amount of air now fills a bigger space, so the pressure inside the lungs drops below outside pressure and air rushes in. (Lungs do not "pull" air like a magnet.)
- 52. (B)** Earthworms breathe through moist skin using oxygen from soil air. Heavy rain fills the soil spaces with water, leaving too little air, so the worms surface to get oxygen.
- 53. (C)** In the dark a plant cannot photosynthesise, but it still respire: it uses oxygen and releases carbon dioxide.
- 54. (A)** Aerobic respiration gives carbon dioxide + water; yeast fermentation gives alcohol + carbon dioxide. The product common to both is carbon dioxide.
- 55. (B)** The hairs and sticky mucus in the nasal cavity trap dust and germs, cleaning (and warming and moistening) the air before it reaches the lungs.
- 56. (C)** On exhalation the diaphragm relaxes and the chest cavity shrinks; squeezing the air into a smaller space raises the pressure inside the lungs above outside pressure, so air flows out.
- 57. (B)** The seeds used up the oxygen (candle goes out) and released carbon dioxide (limewater turns milky) — the signs of living seeds respiring, not photosynthesising.
- 58. (B)** The correct comparison is "a fish uses gills while a cockroach uses tracheae." The other options swap the breathing organs between animals.
- 59. (A)** Both a bicycle pump and the breathing system work by changing the volume of a space, which changes the pressure and so moves air in and out. (The pump does not burn glucose; that part is unique to respiration.)
- 60. (D)** Every living cell keeps respiring even during sleep, so it needs a constant oxygen supply — which is why breathing never stops, awake or asleep.